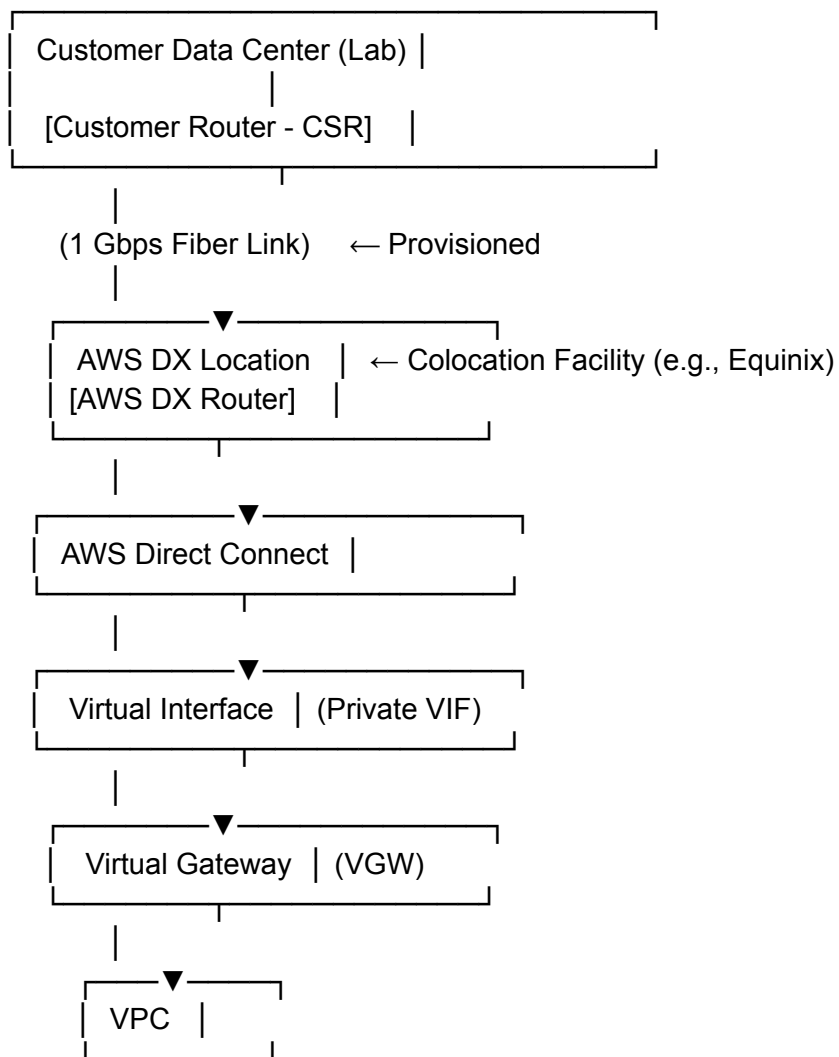

Hands-On Lab: AWS Direct Connect to Access a VPC

Lab Goal

Establish a **private, high-throughput connection** from an **on-premises network (simulated using a router in a colocation facility)** to a **VPC in AWS** using **AWS Direct Connect** with a **Private Virtual Interface (VIF)**.

Lab Architecture Diagram





Lab Prerequisites

1. **AWS Account**
 2. Access to a **Direct Connect Location** (or use a **DX Partner** if you don't have colo access)
 3. **Cisco CSR1000v or virtual router** for simulation (or use AWS Cloud Router/Simulator for demo)
 4. **BGP knowledge** (BGP peering setup is required)
-



Steps Overview

Part A: Set Up Direct Connect in AWS

Part B: Create Private Virtual Interface

Part C: Configure On-Premises Router (CSR)

Part D: Connect to VPC and Validate



Part A: Create a Direct Connect Connection

1. **Go to AWS Console → Direct Connect**
2. Choose **Connections → Create Connection**
 - **Location:** Select AWS DX location (e.g., Singapore Equinix SG1)
 - **Bandwidth:** 1 Gbps
 - **Connection Type:** Dedicated
 - **Name:** MyDXConnection

3. Submit request → You will receive a **LOA-CFA (Letter of Authorization and Connecting Facility Assignment)**

 *Note: This LOA is used by your network provider to complete the cross-connect to AWS.*

✓ **Part B: Create a Private Virtual Interface (VIF)**

1. After the physical connection is **provisioned & active**, go to:
 - **Direct Connect → Virtual Interfaces → Create Virtual Interface**
 2. Choose **Private Virtual Interface**
 3. Set details:
 - **Interface Name:** **DX-to-VPC**
 - **Connection:** Select your DX connection
 - **Virtual Gateway:** Attach to your **VGW** or **Transit Gateway**
 - **BGP ASN:** Your on-prem router's ASN (e.g., 65001)
 - **Your IPv4 CIDR:** e.g., **169.254.100.2/30**
 - **AWS IPv4 CIDR:** e.g., **169.254.100.1/30**
 - **BGP Auth Key** (optional): Password for BGP
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✓ **Part C: Simulate On-Premises Router (CSR) Configuration**

You can use a Cisco CSR1000v in AWS or a physical router.

Example configuration:

```
interface GigabitEthernet1
ip address 169.254.100.2 255.255.255.252
no shutdown
```

```
router bgp 65001
bgp log-neighbor-changes
neighbor 169.254.100.1 remote-as 64512
neighbor 169.254.100.1 password mybgpkey
```

 Replace ASNs and IPs as per your config. Make sure your device can support BGP and IP connectivity.

✓ Part D: Attach Virtual Gateway to VPC

1. Go to **VPC Dashboard** → **Virtual Private Gateway**
 - Create and name a new VGW
 - Attach it to your VPC
 2. The Private VIF you created earlier should be associated with this VGW
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✓ Part E: Route Traffic

1. Modify the **VPC route table**:
 - Destination: Your on-prem network (e.g., **10.0.0.0/16**)
 - Target: Virtual Private Gateway
 2. Ensure **on-prem router** has route back to VPC CIDR
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Validation Steps

Task	Command
Check BGP session	<code>show ip bgp summary</code> on router

Check routes	<code>show ip route</code> or ping from EC2 to on-prem
Ping test	From EC2 → On-Premises or vice versa
Monitor DX	Use CloudWatch metrics for <code>ConnectionState</code> , <code>BpsIn</code> , <code>BpsOut</code>

Optional: Redundancy Setup

For HA, repeat setup using:

- Another **DX location**
 - A **VPN fallback** using Site-to-Site VPN over internet
-

Summary

Component	Description
DX Connection	Dedicated line to AWS
Private VIF	Secure access to VPC
VGW	Gateway for DX/VPN
BGP	Required for routing
CSR Router	Simulated on-prem router
Route Table	Updated for communication
